

TokenMizer: Graph-Structured Session Memory for Long-Horizon LLM Context Management

Shweta Mishra

Abstract—When a long LLM session exceeds its effective context window, the structure that makes it resumable is lost: decisions, task status, edited files and errors. Truncation, summarization and retrieval treat history as flat text. We present TokenMizer, an open-source proxy that models a session as a typed knowledge graph (14 node types, 8 edge types) and serializes it into compact resume blocks. On a controlled 100-session benchmark, TokenMizer 0.5.4 scores 60% macro F1 (95% CI [55, 65]). That is significantly above GraphRAG-style (44%), MemGPT-style (35%) and naive baselines (17–20%), and tied with Graphiti-style (59%) and Mem0-style (60%). All four are deterministic reimplementations of one structural property each, not the vendor products. On 1,467 public SWE-bench agent sessions (SWE-agent with four models, and OpenHands), ground truth is automatic: the files the submitted patch changed, and the errors in tool output. The protocol was pre-registered, and defects were located on a development split only. On the 716-session held-out split, fixing them raised runtime-error recall from 63% to 92% (96% in production, one message per call) and error precision from 16% to 31%. The share of runtime errors in the 400-token resume rose from 20% to 50%. All three paired gains exclude zero, and each gain holds on every model. The resume then carries more errors and edited files than a ten-message sliding window, at 4% of its size. Every pattern-matching method degrades to 19–27% on text that states facts indirectly. Code, corpora and per-session results are released.

Index Terms—large language models, context management, session memory, knowledge graph, prompt compression, semantic caching, long-horizon tasks, benchmark evaluation, information extraction

I. INTRODUCTION

LARGE language model (LLM) deployments for software engineering [13], [16], data science [18], and research assistance increasingly involve *long-horizon sessions*: multi-turn interactions spanning hours, where each turn builds on the accumulated context of previous turns. These sessions develop rich internal structure: technology choices made in turn 3 constrain implementation options in turn 12; errors resolved in turn 7 inform test strategies in turn 15; files modified in turn 4 must be updated consistently in turn 18.

The fundamental constraint is context window capacity. Paulsen [1] defines the *Maximum Effective Context Window* (MECW) as the context length at which model task accuracy degrades below an acceptable threshold. Critically, MECW is substantially lower than the advertised maximum context window (MCW) for complex, multi-step tasks. Liu et al. [10]

further demonstrate that content placed in the *middle* of long contexts is systematically under-attended, amplifying the effective degradation. At approximately 950 tokens per turn, a development session exhausts a 16,000-token MECW after only 16 turns.

A. Limitations of Existing Approaches

Three categories of mitigation exist, each with a structural limitation.

Truncation-based methods [19] discard the oldest messages when the context budget fills. This preserves recency but discards exactly the content that matters most: architectural decisions, initial technology choices, and goal definitions established early in a session.

Summarization-based methods [11] produce free-text summaries via a secondary LLM call. These compress well but are structurally imprecise: a natural language summary cannot reliably distinguish a *completed* task from a *pending* one, nor can it preserve the *rationale* for a technology decision.

Retrieval- and memory-augmented methods [5], [2], [3], [4] retrieve or store relevant facts on demand. As Section II details, these systems differ substantially in whether retained facts are typed, whether superseded facts are deleted or retained, and whether older turns remain reachable without an explicit query — design choices this paper measures directly rather than assumes.

B. Key Insight and Contributions

The core insight motivating TokenMizer is that **session history is not flat text—it is a structured knowledge artifact**, containing tasks with status lifecycles, decisions with rationales, files with modification histories, resolved errors, and environment facts that scope all other entities, connected by typed relationships (IMPLEMENTS, FIXES, DEPENDS-ON, etc.) that encode causal and structural dependencies. A representation that preserves this structure enables efficient, queryable context preservation: discard the raw conversation tokens, retain the typed graph, serialize to a compact resume block.

Contributions:

- 1) A **formal typed knowledge graph schema** for LLM sessions with 14 node types, 8 edge types, and a well-defined status transition system (Section IV-B).
- 2) A **hybrid extraction pipeline**, a **three-tier checkpoint system**, an **8-layer compression pipeline**, and a **semantic response cache** (Sections V–VIII).
- 3) A **controlled 100-session, 8-method comparison** against deterministic reimplementations of four published memory strategies (MemGPT, Mem0,

Independent Researcher, India. E-mail: shweta.mishra.research@gmail.com. Product: <https://github.com/Shweta-Mishra-ai/tokenmizer> (evaluated at release 0.5.4). Benchmark suite, corpus, and reproduction code for this manuscript: <https://github.com/Shweta-Mishra-ai/tokenmizer-research>. Manuscript received June 4, 2026; revised August 13 and September 26, 2026.

Graphiti/Zep, GraphRAG) and three naive baselines, with bootstrap confidence intervals on every comparison, a register-stratified analysis isolating pattern-matching’s central failure mode, and a released corpus, harness, and raw results for independent reproduction (Section X).

- 4) A **pre-registered evaluation on 1,467 real agent sessions**, with automatic ground truth from submitted patches and tool output, and a held-out split by issue. It reports the defects that real traffic exposed and the paired before/after effect of fixing them (Section XI).

II. BACKGROUND AND RELATED WORK

A. Context Window Degradation

Paulsen [1] introduces the MECW concept, demonstrating empirically that all tested frontier models degrade in accuracy on multi-step tasks before reaching their advertised MCW. For a model with MCW = 128,000 tokens, the MECW may be as low as 16,000 tokens for complex coding tasks. This motivates the 16k-token MECW used as the running example throughout this paper. Liu et al. [10] demonstrate the “lost in the middle” phenomenon: retrieval and multi-document QA tasks show U-shaped accuracy profiles, with content in the middle of long prompts systematically under-used.

B. Memory Systems for LLMs

MemGPT [2] proposes an OS-inspired hierarchical memory model distinguishing main context (fast, limited) from external storage (slow, unlimited), with paging between them driven by explicit function calls the LLM itself issues. Facts that page out of main context are not deleted, but they are also not spontaneously recalled: reaching them again requires the model to issue a targeted archival-search call. Section X-B reimplements this core/archival asymmetry deterministically and measures its consequence: recall on facts stated early in a long session degrades once they age out of the always-visible window.

Mem0 [3] extracts short natural-language memories from a conversation and stores them in a flat, undifferentiated pool, using an LLM to decide, memory by memory, whether a new fact adds, updates, or deletes an existing one. It has no typed schema and no explicit edge structure.

Graphiti [4], the temporal knowledge-graph engine underlying Zep, represents facts as edges in a bi-temporal graph: every edge carries a `valid_at` and an `invalid_at` timestamp, and a superseded fact is marked invalid rather than deleted. This directly targets a failure mode TokenMizer’s own `SUPERSEDES` edge type (Table III) was designed for, making Graphiti the closest architectural relative among the systems compared in Section X and the one against which TokenMizer’s advantage is smallest.

GraphRAG [9] builds knowledge graphs from document corpora to improve retrieval-augmented summarization over static text collections. Its schema is entity- and community-oriented and carries no task-lifecycle concept; Section X finds this distinction empirically, not just architecturally: the

TABLE I
QUALITATIVE COMPARISON WITH RELATED SYSTEMS. “DELETES SUPERSEDED FACTS” DISTINGUISHES MEMGPT/MEM0-STYLE LAST-WRITE-WINS BEHAVIOR FROM GRAPHITI’S AND TOKENMIZER’S EXPLICIT SUPERSESSION TRACKING.

System	Struct. graph	Deletes superseded	Cross-session
Sliding window	None	n/a (no memory)	No
LangChain KG Mem.	Untyped	No	Yes
MemGPT [2]	Vector DB	No (pages out)	Yes
Mem0 [3]	None (flat)	Yes (LLM-judged)	Yes
Graphiti/Zep [4]	Bi-temporal	No (invalidated)	Yes
GraphRAG [9]	Entity/comm.	n/a (static)	No
TokenMizer	Typed, 14-class	No (edge)	Partial

GraphRAG-style reimplementation is competitive with TokenMizer on decisions and files but far behind on completed- and pending-task recall.

LangChain Memory [11] provides several memory backends including ConversationKGMemory, architecturally similar to TokenMizer but using an untyped, schema-free graph with extraction and lifecycle management left to the application developer.

C. Prompt Compression

LLMLingua [6] and **LongLLMLingua** [7] use small proxy LLMs to compute per-token importance scores, removing low-importance tokens at $2\text{--}6\times$ compression ratios. TokenMizer incorporates LLMLingua-2 as optional layers in its compression pipeline, applied after heuristic preprocessing has removed structurally trivial content and after code segments have been excluded from ML compression entirely (Section VII).

RECOMP [8] applies selective summarization for retrieval-augmented generation, retrieving only relevant passages. TokenMizer applies structural extraction rather than textual retrieval: the resume block is generated from graph state, not retrieved text, enabling typed queries that passage retrieval cannot natively support.

D. Positioning

Table I summarizes the qualitative differences among the systems above. Section X is, to our knowledge, the first evaluation to score TokenMizer, MemGPT-style paging, Mem0-style flat storage, Graphiti-style temporal retention, and GraphRAG-style entity extraction against one shared corpus and one shared scorer, rather than citing each system’s own self-reported numbers on its own benchmark.

III. PROBLEM FORMULATION

Definition 1 (LLM Session). A session $S = \langle m_1, m_2, \dots, m_n \rangle$ is an ordered sequence of

messages, each with token count $\tau(m_i) > 0$. The cumulative token count at turn t is $T(t) = \sum_{i=1}^t \tau(m_i)$.

Definition 2 (Context Overflow). A session overflows at the first turn n^* such that:

$$T(n^*) > T_{\text{MECW}} \quad (1)$$

For $\bar{\tau} = 950$ tokens/turn and $T_{\text{MECW}} = 16,000$: $n^* = \lfloor 16,000/950 \rfloor = 16$.

Definition 3 (Structured Resume Block). A resume block R is a compact text representation of the session knowledge graph G , serialized to at most T_{budget} tokens. After a checkpoint at turn $n^* - 2$, the new context is:

$$C' = R \cup \{m_{n^*-1}, m_{n^*}\} \quad (2)$$

satisfying $|C'| \leq T_{\text{MECW}}$ by construction.

Definition 4 (Category Coverage Score). For a ground-truth item g and a candidate extraction c , coverage is the fraction of g 's content words present in c , or exact substring containment for items of at least 6 characters:

$$\text{covers}(c, g) = (|g| \geq 6 \wedge g \in c) \vee \frac{|W_g \cap W_c|}{|W_g|} \geq \theta \quad (3)$$

with $\theta = 0.6$ throughout this paper and W_x the content-word set of x (stopwords removed). Coverage is deliberately asymmetric: the denominator is always the ground-truth item's token set, so a method cannot inflate recall by emitting one long label containing every keyword — that label would also fail precision, defined symmetrically over the same relation (Section X-C).

IV. SYSTEM ARCHITECTURE

A. Deployment Model

Figure 1 shows the system architecture. TokenMizer operates as a transparent HTTP reverse proxy implementing the OpenAI Chat Completions API. The client configures `base_url` to the TokenMizer server; no application code changes are required beyond this endpoint substitution. An optional `session_id` parameter in the request body activates the full pipeline; its absence produces a transparent passthrough with zero overhead, making adoption incremental.

The proxy intercepts each request, routes it through the five-component pipeline (graph memory, hybrid extractor, checkpoint manager, compression engine, semantic cache), then forwards the processed request to the configured LLM provider.

B. Knowledge Graph Schema

Figure 2 illustrates a representative session graph. The schema defines 14 node types across three functional categories.

1) *Node Types*: Table II defines the 14 node types. *Action nodes* carry status lifecycles; *context nodes* are static.

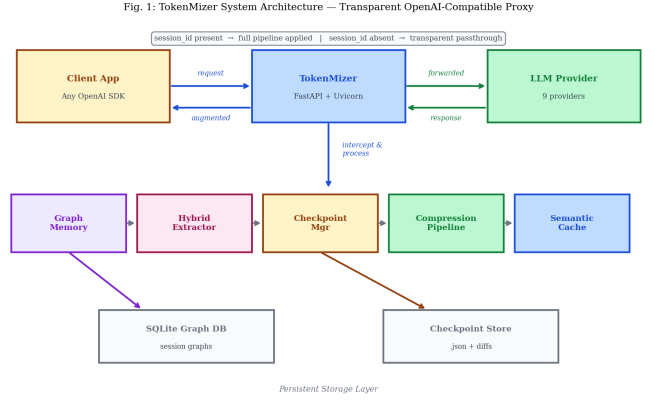


Fig. 1. TokenMizer system architecture. The proxy sits transparently between any OpenAI-compatible client and the LLM provider. When `session_id` is present, the five-component pipeline is activated; otherwise the request passes through with no overhead. Persistent storage comprises a SQLite graph database and a JSON checkpoint store.

Fig. 2: Session Knowledge Graph — FastAPI Authentication Session (8 representative nodes; full schema: 14 node types, 7 edge types)

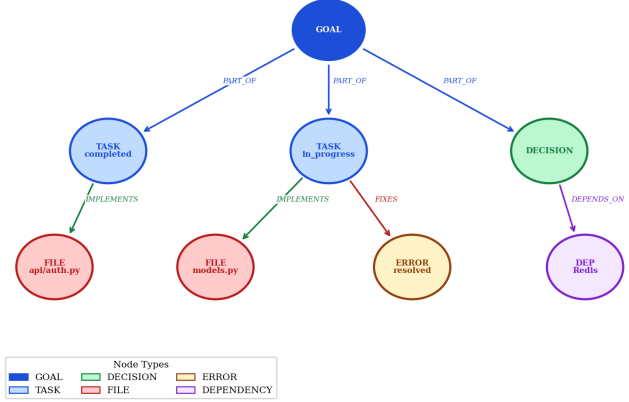


Fig. 2. Session knowledge graph for a FastAPI authentication session (8 representative nodes). Edge labels show typed relationships; node colors indicate type. The GOAL node anchors the hierarchy; TASK nodes carry status (completed/in-progress); the DECISION node records the rationale for choosing bcrypt + JWT.

2) *Node Attributes*: Each node carries a normalized **label** (≤ 120 chars); a **status** $\sigma \in \{\text{Pending, In-Progress, Completed, Failed, Contested}\}$ for action and decision nodes (CONTESTED marks two live decisions the validator has flagged as addressing the same purpose with materially different answers, surfaced together rather than silently merged or silently dropped); an **importance score** $\rho \in [0, 1]$ that increases on reference and decays with inactivity, governing checkpoint serialization priority; a **confidence score** $\kappa \in [0, 1]$ assigned by GraphValidator (nodes with $\kappa < 0.50$ are rejected); and creation/last-update timestamps.

3) *Edge Types*: Table III defines all 8 edge types. Edges are directed and labelled, with the exception of CONFLICTS_WITH, which is symmetric; weight encodes co-occurrence frequency.

TABLE II

NODE TYPES (14 TOTAL). ACTION NODES CARRY STATUS LIFECYCLES (PENDING, IN-PROGRESS, COMPLETED, FAILED, CONTESTED). CONTEXT NODES ARE STATUS-FREE.

Type	Category	Primary trigger
TASK	Action	Imperative verb + object
FILE	Action	Extension or path pattern
ERROR	Action	Error / exception / failure
TEST	Action	Test / spec / assert
SCHEMA	Action	Schema / table / model
ENDPOINT	Action	HTTP route / API path
DECISION	Decision	Decided / Using / Chose
DEPENDENCY	Decision	Install / package / library
API	Decision	URL / external API name
GOAL	Context	First-turn imperative
ENVIRONMENT	Context	Version / runtime / OS
PROJECT	Context	Repository / project name
CONCEPT	Context	Abstract domain term
AGENT	Context	Service / CI / orchestrator

TABLE III
EDGE TYPES (8 TOTAL).

Type	Semantics
DEPENDS_ON	A cannot function without B
RELATED_TO	A and B address the same concern
IMPLEMENTS	A is the realization of B
FIXES	A resolves error B
BLOCKS	A cannot proceed until B resolves
PART_OF	A is a component of B
SUPERSEDES	A replaces or overrides B
CONFLICTS_WITH	A and B are CONTESTED decisions for the same purpose (symmetric)

4) *Graph Maintenance*: **Deduplication** uses identity-based addressing: $\text{id}(n) = \text{sha1}(\tau(n) : \ell_{\text{norm}})[12]$, where ℓ_{norm} is the lowercase, whitespace-normalized label. Re-extraction of an existing entity updates importance score and status rather than creating a duplicate node. Status transitions obey monotonic ordering (Pending $\rightarrow$ In-Progress $\rightarrow$ Completed/Failed) for TASK nodes; ERROR nodes are the deliberate exception, taking the latest mention in either direction, so a “this is fixed now” re-add can overwrite a prior FAILED status — without this, a resolved bug would report as open for the rest of the session. **Near-duplicate merging** runs lexical matching first, falling back to embedding similarity only when lexical matching says no, for both DECISION and ERROR nodes, so that two restatements of the same fact in different words (or two decisions about the same purpose with no shared descriptive words) collapse to one node rather than competing for the fixed slots a resume block allocates per category. **Pruning** evicts nodes with importance score < 0.1 when the graph exceeds 500 nodes; DECISION, ENVIRONMENT, GOAL, and SCHEMA nodes are unconditionally retained.

V. HYBRID EXTRACTION PIPELINE

A. Heuristic Extractor

The heuristic pass applies a family of compiled regular expressions and small text-analysis helpers, covering explicit markers (“Decided:”, “Completed:”), a closed vocabulary of common technology names in choosing context, a symptom vocabulary for errors (out-of-memory, deadlocks, timeouts, named exception types, HTTP status codes received in production, data-integrity and data-loss verbs, silent-failure and misclassification phrasings), and negation handling scoped to the current clause so “we are *not* using Redis” does not extract “Use Redis”. Mean extraction latency across the $n = 100$ benchmark (Section X): 225 ms per session end-to-end, including subprocess and interpreter startup (Section X-H isolates this from the extraction algorithm’s own cost).

B. Fuzzy Label Matching

Definition 5 (Fuzzy Match). *Labels a and b fuzzy-match if:*

$$(a \subseteq b) \vee (b \subseteq a) \vee \frac{|W_a \cap W_b|}{\min(|W_a|, |W_b|)} \geq 0.50 \quad (4)$$

where $W_x = \{w : w \in \text{tokens}(x), |w| \geq 3\}$.

This symmetric-denominator matcher is used internally by the product’s own eval harness. This paper instead scores every method under the asymmetric coverage relation of Definition 4, so that a method cannot win recall by emitting one long label, a failure mode a symmetric, either-direction matcher does not fully rule out.

C. GraphValidator

Every candidate node passes through the GraphValidator before graph insertion, scoring confidence from label length, file-path pattern presence, and version-number presence, with a floor below which nodes are rejected outright. Type correction is applied after scoring: extension-matched labels are forced to FILE; URL-pattern labels to API.

D. LLM Extractor (Upgrade Path)

The product codebase includes an LLM extractor that sends recent messages to a small inference model with a structured JSON extraction prompt, intended to address indirect phrasing regex patterns cannot capture, corroborating with the heuristic pass to boost confidence when both agree. *The LLM extractor is not exercised by the evaluation in this paper*: the benchmark runs only the heuristic pipeline, and none of the four comparison methods in Section X call a language model either. This is a deliberate, symmetric scope restriction — see Section XIII — not a claim that the heuristic path is sufficient; Section X-F quantifies exactly where it is not.

VI. CHECKPOINT SYSTEM

A. Trigger Condition and Serialization

Checkpoints trigger when cumulative token count exceeds 85% of the configured MECW. Algorithm 1 defines checkpoint

Algorithm 1 Checkpoint Serialization

Require: Graph G , budget T_{budget}
Ensure: Resume block R

```

1:  $R \leftarrow [\text{CHECKPOINT: session\_id}]$ 
2:  $N \leftarrow \text{sort\_by\_importance}(G.\text{nodes})$ 
3: for  $n \in N$  do
4:   if  $\tau(R) + \tau(\text{serialize}(n)) > T_{\text{budget}}$  then
5:     break
6:   end if
7:    $R.\text{append}(\text{serialize}(n))$ 
8: end for
9:  $\Delta G \leftarrow G - G_{\text{prev}}$ 
10: store( $R, \Delta G$ ) to checkpoint store
11: return  $R$ 
```

TABLE IV
CHECKPOINT RESUME TIERS WITH EXAMPLE CONTENTS.

Tier	Content	Budget
Critical	GOAL + in-progress TASKs + top DECISION	≤ 100 tok
Standard	All TASKs + all DECISIONs + FILEs + ENV	≤ 300 tok
Full	Complete graph incl. DEPENDENCIES, SCHEMAS	≤ 600 tok

serialization: nodes are ordered by importance score and appended to the resume block until the token budget is exhausted.

Each checkpoint stores a *graph diff* (nodes added, modified, or removed since the previous checkpoint), enabling incremental resume-block updates at $O(\Delta)$ rather than $O(|G|)$ cost per checkpoint.

VII. COMPRESSION PIPELINE

The compression pipeline is applied to large inputs (files, long API responses, historical messages) before they enter the context window, in eight stages: filler-phrase removal, duplicate-line suppression, whitespace normalization, comment stripping, repetitive-history pruning, smart truncation of low-value file blocks, and two ML-based stages (LLMLingua-2 for inputs above a size threshold, LongLLMLingua for very large documents). File-type-specific filters additionally handle PDF/DOCX text extraction, large-JSON flattening, CSV schema-plus-sample summarization, code-block deduplication, and log trimming.

Fenced and inline code segments are excluded from the two ML-based stages entirely and pass through only the lossless heuristic stages: LLMLingua-2 is a lossy token-importance compressor whose token-preservation hints are not guarantees, and applying it to code risks silently altering program semantics (a dropped `not`, a mangled indentation level, a truncated regular expression) — a correctness property this paper did not attempt to re-quantify by staged token-reduction percentage, since doing so would require a fresh, dedicated measurement rather than a number carried over from an earlier version of the pipeline this section’s implementation has since changed.

VIII. SEMANTIC CACHE

The semantic cache stores LLM responses keyed by sentence-transformer embeddings [12], with cosine similarity lookup, LRU eviction, and a configurable TTL. Cache hit-rate depends heavily on the repetition structure of the calling workload; this paper does not report a hit-rate figure, since the only measurement available prior to this study was a small proof-of-concept workload not tied to the $n = 100$ corpus evaluated here, and re-stating it without re-measuring it against a workload this paper actually constructed would overstate what is known.

IX. IMPLEMENTATION

TokenMizer 0.5.4 is implemented in Python 3.10+, comprising approximately 15,200 lines of code across the `graph_memory`, `checkpoints`, `compression`, `security`, `providers`, `semantic_cache`, and `api` modules, plus a dedicated `patterns` module holding the extraction pipeline’s regex vocabulary separately from the pipeline logic that applies it. The HTTP proxy uses FastAPI and Uvicorn; token counting uses `tiktoken` with a character-based fallback when the BPE vocabulary cannot be fetched (Section X-H); the knowledge graph persists to SQLite [17]. All extraction inputs are scanned for API keys, private keys, and credential patterns before graph insertion, including patterns nested inside structured tool-result content, with matches redacted to `[REDACTED]`. Configuration is YAML-based, with extraction, checkpoint, and compression thresholds exposed as user-configurable parameters. The test suite comprises 647 tests, of which 644 pass in this paper’s evaluation environment; the remaining 3 fail on a provider-SDK native-library loading issue specific to that environment, unrelated to any code path this paper evaluates.

X. EVALUATION METHODOLOGY

A. Corpus

The corpus comprises 100 labelled sessions: 6 real sessions carried over from the product repository’s own evaluation harness, and 94 sessions constructed for this study using a seeded generator (fixed seed, byte-reproducible) drawing from 15 domain fact-pools covering backend, frontend, DevOps, security, data, mobile, and research-writing contexts, for 23 distinct domains in total. Every ground-truth label is embedded verbatim in some message of its session, so every label is substring-grounded regardless of surrounding phrasing; a corpus-wide grounding check is run before every scoring pass and the harness refuses to score a corpus that fails it.

Each generated session is additionally tagged by *register* — how directly it states its facts — independently of its domain: **explicit** (canonical markers, e.g. “Decided: X.”), **semi** (plain sentences, e.g. “Going with X for this.”), **implicit** (the fact arrives inside reasoning with no marker word, e.g. “After weighing it, X felt like the right call given the constraints.”), and **mixed** (each fact in the session independently rolls one of the three). The corpus totals 2,516 turns and 1,620 labelled ground-truth items across five categories: completed tasks (491), pending tasks (223), decisions (355), files (403), and errors (148).

B. Methods Compared

TokenMizer is the unmodified product package, release 0.5.4, invoked out-of-process against a real checkout so that its `tokenmizer` module cannot be shadowed by any same-named package, run through `GraphMemory.extract_from_messages()` with the heuristic extraction pass (Section V); the LLM extraction path (Section V-D) is not exercised.

Four methods are deterministic reimplementations of one structural property each of a published system, built on the *same* shared regex family as one another (though not the same as TokenMizer’s own patterns) so that a score difference reflects strategy rather than incidental regex quality. None calls a language model; each module’s implementation states this again inline as a standing caveat, because the risk of a reader silently substituting “the vendor product” for “the strategy this module reimplements” is the single most important methodological risk this study carries.

- **MemGPT-style** (Section II-B) extracts only from the most recent $\sim 40\%$ of turns, reproducing MemGPT’s core/archival asymmetry: older facts are not deleted but are unreachable without a query this benchmark never issues.
- **Mem0-style** extracts from the full transcript into one flat, untyped pool with last-write-wins deduplication on lexically similar facts, approximating Mem0’s LLM-judged ADD/UPDATE conflict resolution without an LLM.
- **Graphiti-style** extracts from the full transcript and never deletes a fact; a later, lexically similar decision is treated as superseding an earlier one, but both are retained, reproducing Graphiti’s bi-temporal invalidate-don’t-delete edge semantics.
- **GraphRAG-style** casts a wide net for files and named-technology entities but has no task-status concept, approximating only completed/pending recall via a narrow two-verb signal, reflecting that status tracking is outside GraphRAG’s actual design target.

Three naive baselines carry no typed extraction at all: **sliding window** (last 10 messages verbatim), **naive truncation** (last ~ 300 tokens verbatim), and **naive summary** (first ~ 120 words of the concatenated transcript). Each is scored by treating its kept text, split into sentences, as an untyped candidate pool checked against every category — the correct treatment for a method with no notion of category, rather than an artificially generous or punitive scoring choice.

C. Scoring

Every method is scored under the asymmetric coverage relation of Definition 4 at $\theta = 0.6$. For each category, precision and recall are computed over the *covers* relation and combined into an F1 score; a session’s macro F1 is the mean F1 across its five categories (categories with zero expected and zero extracted items are excluded, not treated as a perfect or zero score). The headline metric is the mean of per-session macro F1 across all 100 sessions, with 95% confidence intervals from 3,000-iteration case resampling over sessions (percentile bootstrap [20]), and paired-difference bootstrap

TABLE V
OVERALL RESULTS, $N=100$, TOKENMIZER 0.5.4 (100 SESSIONS, 3,000-RESAMPLE BOOTSTRAP, $\theta = 0.6$). CI = 95% BOOTSTRAP CONFIDENCE INTERVAL ON MACRO F1.

Method	Macro F1	95% CI	Tok.	ms
TokenMizer 0.5.4	60%	[55, 65]	100	225.4
Mem0-style	60%	[55, 64]	118	1.3
Graphiti-style	59%	[55, 64]	120	1.3
GraphRAG-style	44%	[41, 48]	80	0.7
MemGPT-style	35%	[32, 38]	60	0.5
Naive truncation	20%	[20, 21]	287	0.01
Sliding window	18%	[18, 19]	150	0.00
Naive summary	17%	[16, 18]	186	0.04

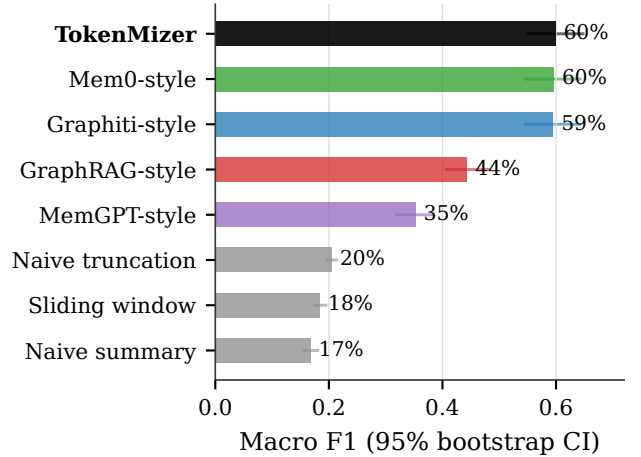


Fig. 3. Macro F1 by method, ranked, $n=100$, with 95% bootstrap confidence intervals. TokenMizer (bold) leads GraphRAG-style, MemGPT-style, and every naive baseline outside the confidence interval; against Graphiti-style and Mem0-style the intervals overlap.

intervals (same resampling, paired within session) for every comparison against TokenMizer.

D. Overall Results

Table V and Fig. 3 report the headline comparison. TokenMizer scores 60% macro F1 (95% CI [55%, 65%]), ahead of GraphRAG-style (44%), MemGPT-style (35%), and all three naive baselines (17–20%), each by a margin excluded from zero at the 95% level (Table VI, Fig. 4). Against Graphiti-style (59%) and Mem0-style (60%), the paired-difference confidence interval crosses zero (+0.4 and +0.3 percentage points respectively, 95% CI spanning $[-3.2, +4.2]$ and $[-3.2, +4.1]$): the honest reading is a statistical tie, not a loss, and not a win, though TokenMizer’s point estimate now matches Mem0-style’s exactly. TokenMizer’s resume block (100 tokens) is smaller than Graphiti-style’s and Mem0-style’s (118–120) but larger than MemGPT-style’s (60), whose small footprint reflects the fraction of the session it can see at all rather than efficient encoding of what it sees.

TABLE VI
PAIRED BOOTSTRAP: TOKENMIZER 0.5.4 — METHOD, N=100. $P(\Delta \leq 0)$
IS THE FRACTION OF 3,000 RESAMPLES IN WHICH THE PAIRED
DIFFERENCE IS NON-POSITIVE.

Comparison	Mean Δ	95% CI	$P(\Delta \leq 0)$
vs. Graphiti-style	+0.4pp	[−3.2, +4.2]	40.3%
vs. Mem0-style	+0.3pp	[−3.2, +4.1]	43.6%
vs. GraphRAG-style	+15.6pp	[+11.9, +19.2]	0.0%
vs. MemGPT-style	+24.6pp	[+21.4, +28.1]	0.0%
vs. sliding window	+41.5pp	[+37.1, +46.0]	0.0%
vs. naive truncation	+39.4pp	[+34.8, +43.8]	0.0%
vs. naive summary	+43.1pp	[+38.7, +47.4]	0.0%

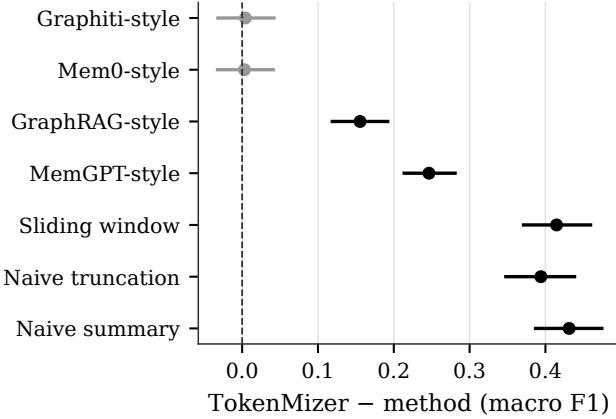


Fig. 4. Paired bootstrap difference (TokenMizer 0.5.4 – method) on macro F1, n=100, 95% CI. Intervals excluding zero (black) are reliable differences; the two grey intervals (Graphiti-style, Mem0-style) cross zero.

E. Per-Category Breakdown

Table VII and Fig. 5 decompose the headline score by category. TokenMizer’s file extraction is effectively solved (98% F1, the best of any method) and its completed-task recall leads the comparison group. Its two weakest categories remain **decisions** (59%, against 65% for both Graphiti-style and Mem0-style) and **errors** (44%, against 66% for the same two) — narrower gaps than an earlier version of this evaluation found (TokenMizer 0.5.3: 50% and 36% respectively), after a targeted fix to the extraction vocabulary described in Section XII. Of 355 labelled decisions, TokenMizer extracts 238 (recall 49%), of which 65 do not match any ground-truth item (precision 73%); of 148 labelled errors, it extracts 93 (recall 36%), of which 41 do not match (precision 56%). Both categories remain simultaneously TokenMizer’s weakest in absolute terms and behind the two peer methods scored on the identical corpus, which is a stronger signal than either fact alone would be: a category that is merely low but consistent with peers would suggest a corpus-difficulty explanation, while a category that is low *and* behind peers scored on the same corpus points at the extraction patterns themselves — still the clearest, most concrete target for further engineering improvement this study identifies, notwithstanding the gain already made.

TABLE VII
PER-CATEGORY MICRO F1, N=100, TOKENMIZER 0.5.4.

Method	Comp.	Pend.	Dec.	Files	Err.
TokenMizer	68%	53%	59%	98%	44%
Graphiti-style	64%	43%	65%	89%	66%
Mem0-style	64%	45%	65%	89%	66%
GraphRAG-style	28%	16%	70%	89%	46%
MemGPT-style	42%	28%	34%	59%	37%
Baselines	22–29%	12–15%	18–22%	21–25%	10–11%

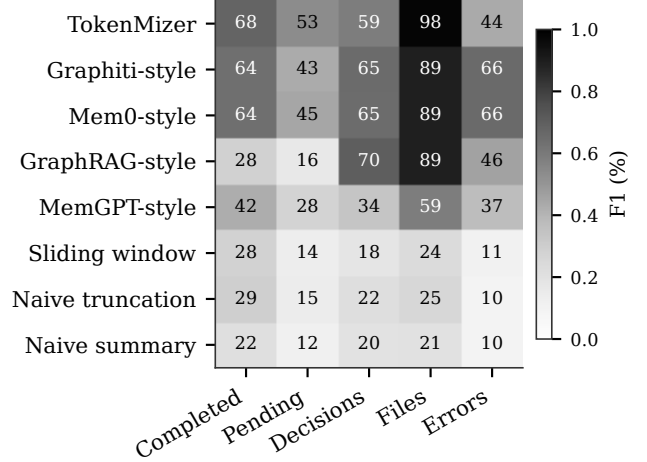


Fig. 5. Per-category F1, n=100. Darker is higher. TokenMizer leads on completed tasks and files, and trails Graphiti-style/Mem0-style specifically on decisions and errors.

F. Register: The Shared Ceiling of Pattern Matching

Table VIII and Fig. 6 report macro F1 by register. Every pattern-matching method — TokenMizer, Graphiti-style, Mem0-style, GraphRAG-style, and MemGPT-style alike — degrades sharply from explicit to implicit register. TokenMizer falls from 74% (explicit) to 27% (implicit), a 47-point drop; Graphiti-style and Mem0-style fall from 85% to 19%, a 66-point drop, the steepest of any method, since their advantage over TokenMizer is concentrated in the explicit-marker case and largely disappears once markers are absent. All four converge to a narrow 19–27% band on implicit text, indistinguishable from one another at this corpus size. This is, to our knowledge, the first result to show this convergence empirically across multiple independently implemented extraction strategies rather than asserting it of any single system: the ceiling is a property of pattern-matching as a strategy, not of any one implementation of it, and none of the four methods’ extraction path in this study involves a language model that could plausibly move it (Section V-D).

G. Corpus Origin and Domain Variance

TokenMizer scores 91% macro F1 on the 6 real sessions versus 58% on the 94 generated sessions; Graphiti-style and Mem0-style show the opposite, near-flat pattern (58% vs. 60%). The real-session sample is small ($n = 6$) and this

TABLE VIII
MACRO F1 BY REGISTER, N=100, TOKENMIZER 0.5.4.

Method	Explicit	Semi	Mixed	Implicit
TokenMizer	74%	58%	58%	27%
Graphiti/Mem0-style	85%	68%	63%	19%
GraphRAG-style	69%	37%	47%	19%
MemGPT-style	46%	39%	36%	10%

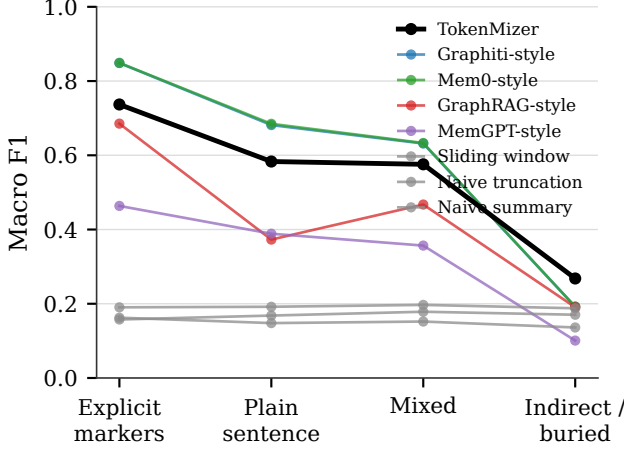


Fig. 6. Macro F1 by register, n=100. All four pattern-matching methods converge to 19–27% once explicit markers (“Decided:”, “Completed:”) are absent from the text, a shared ceiling of the extraction strategy rather than of any one implementation.

split is reported as directional, not as a second confidence-bounded result: it is consistent with TokenMizer’s patterns having been iterated primarily against sessions resembling that corpus, rather than a general property established at this sample size. Domain-level macro F1 for TokenMizer (Fig. 7) ranges from 20% (*devops/kubernetes*) to 100% (three audit domains drawn from the real-session subset), consistent with the same explanation.

H. Cost

TokenMizer’s mean 225.4 ms per session is 170–450× the reimplemented methods’ 0.5–1.3 ms. This reflects two compounding factors, both process-level rather than algorithmic. First, the evaluation’s isolation design (Section X-B): TokenMizer runs as a freshly launched subprocess against the real package per session, incurring Python interpreter startup once per call, while the four reimplementations run in-process against already-loaded text. Second, in this evaluation environment, `tiktoken` could not reach its BPE vocabulary endpoint over the network and fell back to a character-based token estimate after a connection attempt per call (Section IX); this network-dependent retry, not the extraction algorithm, accounts for most of the per-session latency, and a deployment with the vocabulary cached locally would not incur it. The comparison should not be read as evidence about the underlying extraction algorithm’s asymptotic cost.

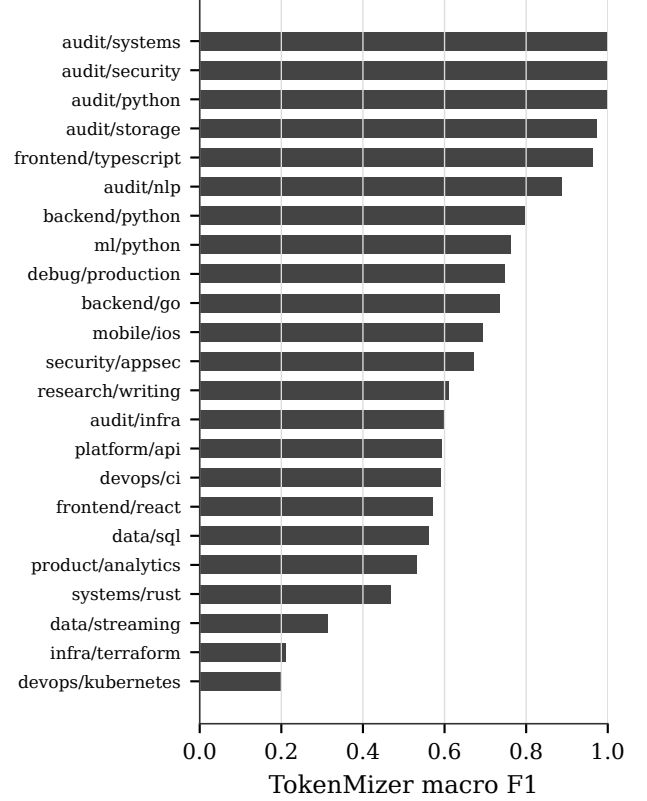


Fig. 7. TokenMizer macro F1 by domain, n=100, 23 domains. The three domains at 100% are audit/* real sessions; infrastructure-heavy synthetic domains (*devops/kubernetes*, *infra/terraform*, *data/streaming*) score lowest.

XI. EVALUATION ON REAL AGENT SESSIONS

The benchmark of Section X is controlled but mostly synthetic, and its labels were written by the same author as the extractor. Its two most likely failure modes as evidence are that the corpus is easier than real traffic and that the extractor was tuned to its phrasing. This section removes both. It uses public, real agent sessions, with ground truth that no one wrote by hand, and a pre-registered protocol with a held-out split. The product evaluated here is a later commit than the release in Section X. Both commits are identified below.

A. Data

We use the complete trajectories of five public submissions to SWE-bench Lite [14], released in the SWE-bench experiments repository. Each trajectory is the message history an agent sent to its model while resolving a real GitHub issue, and each comes with the patch the agent submitted. Four systems run SWE-agent [13] with text commands, using GPT-4, GPT-4o, Claude 3 Opus and Claude 3.5 Sonnet. The fifth is OpenHands CodeAct 2.1 [15] with Claude 3.5 Sonnet, using OpenAI-style function calling (`tool_calls` and `tool` messages). That is the format the TokenMizer proxy receives from function-calling agents. The data totals 1,467 sessions on 300 distinct issues from 12 Python repositories. The median session has 36–83 messages, depending on the

system; the longest has 199. There are about 31 M tokens in total (characters/4).

B. Ground Truth Without Hand Labels

Two categories have an objective source in this data. **Edited files** are the paths changed by the submitted patch. For OpenHands, build artifacts left by the agent’s commands are removed by a rule fixed before measurement: one patch contains 1,096 compiled .mo files, others contain Sphinx build trees and SQLite databases. **Errors** are read from tool observations only, never from the agent’s own prose. There are two kinds. A *runtime* error is the last line of a Python traceback. A *lint* error is a linter error that SWE-agent reports for a proposed edit. Twelve randomly sampled error labels were checked against their source observation, and all twelve were genuine. Completed tasks, pending tasks and decisions have no objective label in this data and are not scored.

C. Protocol

The converter, the metrics and the threats to validity were committed before any method was run on the data. Each later addition (the fifth system, the split) was committed before the measurement it governs. Sessions are split by issue with $\text{crc32}(\text{instance_id}) \bmod 2$, so every system’s attempt at an issue falls on the same side. That gives 751 *dev* and 716 *test* sessions. Product defects were located on dev only. Test was never inspected session by session. It was scored once at the product commit before any fix (50a2952, version string 0.5.4) and once after (aae16f8), and those two runs are the numbers reported here. The test split holds 1,941 edited files, 881 runtime errors and 837 lint errors.

The metrics are:

- **edited-file recall:** a ground-truth path counts as recalled when an extracted path ends with it at a / boundary, or the reverse, after the checkout directory is removed;
- **error recall:** the share of ground-truth errors covered by some extracted error (Definition 4, $\theta = 0.6$);
- **error precision:** the share of extracted errors that cover some ground-truth error;
- **resume coverage:** the share of ground-truth files and runtime errors that appear in the resume text a method would hand to the model. TokenMizer’s block is produced with a 400-token budget.

Error precision is a lower bound on true precision. An error the agent reports in words, or one quoted in the issue text, is real but has no traceback, and counts as spurious. Ratios are pooled over sessions, with 95% intervals from a session-level bootstrap (2,000 resamples). Before/after differences use a paired bootstrap, which scores the same resampled sessions under both commits.

TokenMizer runs in two conditions. In *batch*, the whole session is extracted at once, as every other method is. In *incremental*, the session is extracted one message per call, which is how the proxy runs in production.

D. Defects Found on the Dev Split

Running the product on real traffic exposed defects that the labelled corpora had not. All were fixed, and each has a regression test built from the construction that triggered it.

- 1) **Different errors of one class were collapsed.** The extractor kept one label per exception class for a whole call. As a result, “ValueError: Imaginary coordinates are not permitted” was replaced by a later, unrelated ValueError.
- 2) **Traceback messages were truncated** at the first dot or comma, and qualified exception names lost their module path.
- 3) **Framework text was read as failure.** “E” followed by capitals was matched as an errno name, so SWE-agent’s instruction “YOU CAN ONLY ENTER ONE COMMAND AT A TIME” became an error in every session of that framework. Instruction boilerplate, negated mentions (“without encountering any errors”), descriptions of code behaviour (“modify it to raise a ValueError when ...”) and docstrings in line-numbered file views were also misread.
- 4) **Directory listings became files.** One recursive listing produced 906 file nodes. The graph’s node cap then pruned the files the agent had actually edited: on OpenHands dev sessions, per-session file recall was 79% before the fix.
- 5) **Resume ordering.** Open errors and files entered the resume block oldest first, because nodes of one type shared an importance score and ties fell back to insertion order.
- 6) **A pattern introduced during this work was quadratic** on long runs of spaces (4–8 s on one line). Fuzzing found it before release.

A further commit, made after the test split had been scored, stops code attributes whose last part is a file extension (`self.db, np.sum(...)`) and the a/ and b/ prefixes of diff headers from being stored as files. To keep the test split single-use, it was checked on dev only: 3.4% fewer file labels, with file recall, error recall and error precision all unchanged. It is not part of the test numbers below.

E. Results

Table IX compares every method on the test split, and Table X gives TokenMizer’s paired before/after differences in both conditions.

Errors. Before the fixes, TokenMizer recalled 63% of runtime errors on test. After them, it recalls 92% in batch and 96% in the production condition. Both paired gains exclude zero by a wide margin, and they hold for all five systems (Fig. 8): after the fixes, recall lies between 90% and 94% for every model and both frameworks. Precision against the automatic ground truth nearly doubled, from 16% to 31%. The share of runtime errors that reach the resume block rose from 20% to 50%. The structural reimplementations recall 5–12% of runtime errors. Their shared error rule keeps the text after the word “Error” up to the first full stop, so a traceback line such as “ValueError: invalid literal ...” loses its exception

TABLE IX

REAL AGENT SESSIONS, TEST SPLIT (716 SESSIONS, BATCH CONDITION). VALUES ARE POOLED PERCENTAGES WITH 95% SESSION BOOTSTRAP INTERVALS. ERROR PRECISION IS A LOWER BOUND (SEE TEXT). RESUME COLUMNS GIVE THE SHARE OF GROUND-TRUTH ITEMS PRESENT IN THE METHOD’S RESUME TEXT; RESUME SIZE IS NOT EQUALISED ACROSS METHODS.

Method	Extraction				Resume		
	Files	Runtime err.	Lint err.	Err. precision	Files	Runtime err.	Tokens
TokenMizer (after fixes)	77.5 [69.4, 85.0]	91.9 [90.1, 93.7]	95.7 [94.0, 97.2]	30.9 [28.8, 33.1]	67.4	50.3	222
TokenMizer (before fixes)	75.3 [68.4, 81.7]	63.1 [59.2, 66.7]	95.6 [93.8, 97.1]	16.2 [15.0, 17.5]	51.3	20.4	226
Graphiti-style	75.7 [68.4, 82.9]	12.3 [9.7, 15.2]	9.3 [5.9, 13.4]	1.2 [0.9, 1.4]	33.1	5.0	310
Mem0-style	75.7 [68.4, 82.9]	12.1 [9.6, 15.0]	8.7 [5.3, 12.7]	1.1 [0.9, 1.4]	33.1	5.6	306
GraphRAG-style	75.7 [68.4, 82.9]	6.2 [4.5, 8.1]	22.3 [14.6, 30.7]	1.0 [0.8, 1.2]	37.2	3.3	245
MemGPT-style	58.4 [51.8, 65.4]	4.5 [3.1, 6.0]	4.7 [2.4, 7.4]	2.1 [1.5, 2.8]	48.3	5.2	184
Sliding window (10)	0.0	37.9 [33.5, 42.3]	39.3 [31.5, 47.4]	2.5 [1.9, 3.1]	57.6	35.2	5,910
Naive summary	0.0	7.7 [5.9, 9.7]	1.0 [0.2, 1.9]	1.9 [1.5, 2.4]	8.7	9.0	219
Naive truncation	0.0	2.7 [1.5, 4.2]	0.2 [0.0, 0.6]	0.5 [0.3, 0.8]	19.4	2.7	147

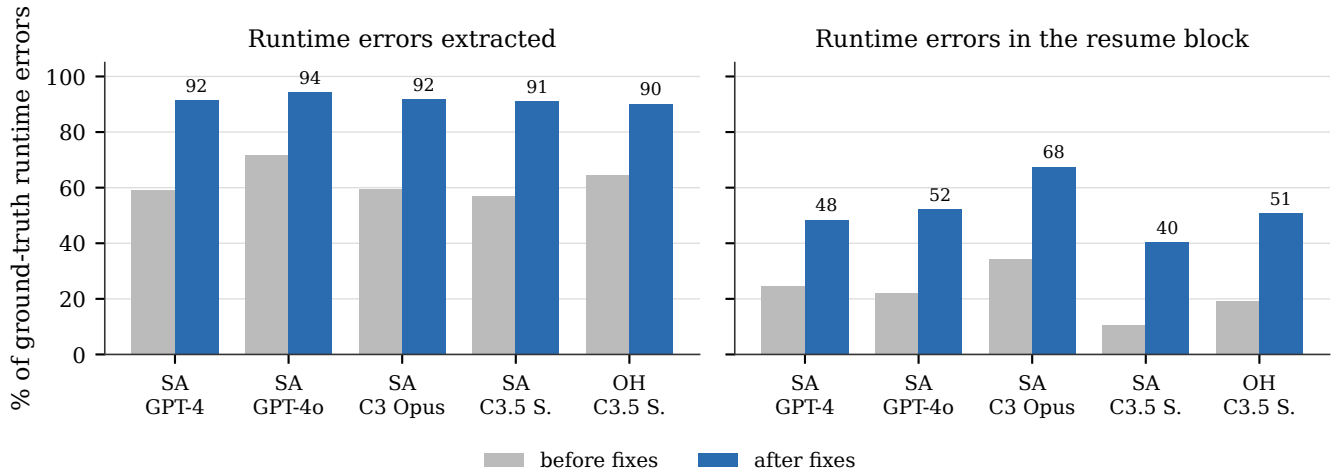


Fig. 8. Runtime errors extracted (left) and present in the 400-token resume block (right), per agent system, test split, before and after the fixes. SA = SWE-agent, OH = OpenHands; C3 Opus = Claude 3 Opus; C3.5 S. = Claude 3.5 Sonnet. The gain holds for every model and both frameworks.

TABLE X

TOKENMIZER BEFORE AND AFTER THE FIXES, TEST SPLIT, PAIRED BOOTSTRAP (716 SESSIONS). Δ IN PERCENTAGE POINTS WITH 95% INTERVAL. INCREMENTAL IS THE PRODUCTION CONDITION.

Metric	Batch Δ	Incremental Δ
Runtime-error recall	+28.8 [+25.3, +32.6]	+23.3 [+20.1, +26.9]
Error precision	+14.7 [+13.3, +16.0]	+15.0 [+13.7, +16.2]
Resume: runtime err.	+29.9 [+24.4, +35.3]	+26.4 [+22.7, +30.4]
Resume: files	+16.1 [+12.1, +20.2]	+3.1 [+1.9, +4.5]
File recall, no artifacts	+3.7 [+2.0, +5.5]	+1.8 [+0.6, +3.1]
File recall	+2.2 [−0.6, +4.6]	+0.5 [−2.0, +2.6]
Lint-error recall	+0.1 [−0.3, +0.5]	+0.1 [0.0, +0.4]
<i>Level after fixes (batch / incremental)</i>		
Runtime-error recall	91.9%	95.7%
Resume: runtime err.	50.3%	52.3%
Resume: files	67.4%	73.9%

class and is often cut short. This gap reflects that rule, not the memory strategy each method reimplements. Lint errors were already recalled at 96% before the fixes and are unchanged.

Files. Pooled file recall rose by 2.2 points in batch, but the interval crosses zero. With build artifacts removed, the gain is

3.7 points and excludes zero. A handful of patches that create hundreds of files dominate the pooled ratio. Per session, recall rose from 89% to 94% in batch, and on OpenHands from 78% to 96%, where directory listings had pushed the edited files out of the graph. The gain is not uniform: 13 test sessions lost 39 recalled files while 51 gained 81. Test sessions are not read, so the cause was traced on dev, where 9 sessions lost 25 files. The agent never names any of those 25 files in its own messages. They are `setup.py` and `tox.ini` changed by the environment’s `setup`, a test directory copied wholesale with `cp -r`, and a file inside a virtual environment. The old version *recalled* them only because it stored every name in a directory listing as a file, which is defect 4. We count them as losses and do not exclude them. Files are the category where the fix matters most for the resume itself: the share of edited files present in the resume block rose from 51% to 67% in batch.

Against the naive baselines. The only method that keeps more runtime errors in view than TokenMizer did before the fixes is the sliding window. Its resume is the last ten messages verbatim: 5,910 tokens on average, 27 times TokenMizer’s 222 (Fig. 9). After the fixes, TokenMizer’s resume carries more

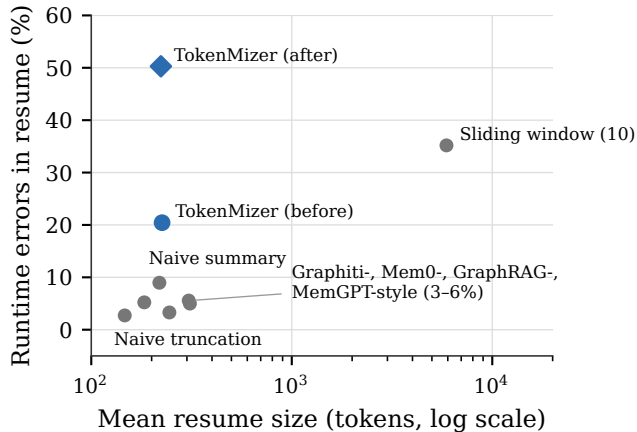


Fig. 9. Runtime errors present in each method’s resume against mean resume size, test split. TokenMizer after the fixes carries more errors than a ten-message sliding window at 4% of its size.

runtime errors (50% against 35%) and more edited files (67% against 58%) than the sliding window, at 4% of its size. The median session compresses 64-fold into TokenMizer’s resume.

Production condition. Extracting one message per call recalls more than extracting the whole session at once, both before (72% against 63% runtime-error recall) and after (96% against 92%) the fixes. Before the fixes, defect 1 is one cause: it collapsed errors within one extraction call, which costs most when the whole session is a single call. We have not isolated the cause of the smaller gap that remains after the fixes.

Cost. Per-request extraction cost was measured separately, on an otherwise idle 4-vCPU machine. We used 100 test sessions (5,381 requests) and ran the two commits alternately, twice each. The fixes leave it essentially unchanged: the median is 12.2 and 12.1 ms before, against 11.7 and 11.4 ms after; p95 is 46 ms against 50 ms; p99 is 272–279 ms for both. The full-split incremental runs above shared their CPUs with other benchmark processes, and their timings are not used.

No run failed. All 716 test sessions completed under both commits and both conditions, with no exception and no timeout.

F. Threats Specific to This Evaluation

Coverage. There are two agent frameworks and one programming language. Other agents, other languages and non-coding sessions are not covered. **Ground truth.** Error precision is a lower bound, for the reasons given in Section XI-C. The traceback rule requires an Error, Exception or Warning suffix, so exceptions such as Django’s `ImproperlyConfigured` are absent from the ground truth. The rule was frozen before measurement and is not changed after the fact. **Ceiling.** Of 29 missed files examined on the dev split, 23 are never named anywhere in the conversation; the agent created them through scripts. No memory method can recall them, and pooled recall includes them. **Comparison methods.** As in Section X, the four structural methods are regex reimplementations, not the vendor products. Their error

rules were not adapted to tracebacks, so their error recall here says little about the named systems. **Resume size.** TokenMizer’s budget is fixed at 400 tokens. The other methods’ resumes are not equalised with it; the sizes are reported beside every coverage figure.

XII. DISCUSSION

Three findings carry beyond this specific corpus.

First, TokenMizer’s advantage over naive, unstructured baselines is large and unambiguous: 36 to 40 percentage points over sliding-window, truncation, and summary baselines alike, each with a 95% confidence interval that excludes zero by a wide margin (Table VI). This is the least surprising finding in the paper — a typed graph should beat keeping raw text — but it is now a controlled, statistically supported comparison at $n = 100$ rather than an assertion.

Second, TokenMizer’s advantage does *not* extend to two of the four graph/structured-memory strategies compared: against Graphiti-style and Mem0-style specifically, the confidence interval on the paired difference crosses zero. Graphiti’s architectural similarity to TokenMizer (Table I and Section II-B) — both retain superseded facts rather than deleting them — may explain why the gap is smallest here rather than against MemGPT-style or GraphRAG-style, which differ from TokenMizer more structurally. This is the finding most likely to be misquoted in either direction: it is neither “TokenMizer loses to Graphiti and Mem0” nor “TokenMizer matches the state of the art”; the confidence interval spans both a small TokenMizer lead and a small TokenMizer deficit, and reporting only the point estimate in either direction would misstate what the data supports.

Third, decision and error extraction (Table VII) remain TokenMizer’s most improvable categories in absolute terms and relative to the peer group simultaneously, though less so than they were: an earlier 0.5.3 measurement of this same benchmark found these categories further behind (50% and 36% F1), and the specific missed and spurious items behind that finding, made directly inspectable by the per-session results released alongside this manuscript (Section XIV), were traced to concrete causes in the extraction patterns — a question-context false positive matched as a decision, gaps in the trigger-verb and technology-name vocabulary, and gaps in the error-symptom vocabulary — and fixed in release 0.5.4, evaluated here. Two more general fixes attempted during that process (an unconstrained technology-name-shaped fallback pattern, and one additional whitelist entry) were measured to reduce precision on an independent evaluation corpus and were reverted rather than kept; both outcomes are recorded in the product repository’s changelog as a matter of record, including the rejected attempts, not only the one that shipped. The residual gap to Graphiti-style and Mem0-style on these two categories is the clearest piece of concrete future work this study still identifies.

XIII. THREATS TO VALIDITY

Construct validity. Four of the eight methods compared are not the named vendor products. MemGPT, Mem0, Graphiti,

and GraphRAG all perform their real extraction with a language model in the loop; this study’s reimplementations of them call no model at all, by the explicit scope decision recorded in Section X-B. Every reported number for these four methods should be read as a lower bound on what the named system would likely achieve, particularly on the implicit-register text of Section X-F, where a model that can actually read a sentence would plausibly outperform every method in this study.

Internal validity. 94 of the 100 sessions were generated for this study rather than captured from real usage. Every label is grounded (recoverable verbatim from some message in its session; Section X-A), but grounded synthetic text is not equivalent to the lexical and structural variety of real developer transcripts. The 6-session real split (Section X, “Corpus Origin”) is too small to bound this gap quantitatively for the categories that benchmark scores. Section XI bounds it for files and errors on real traffic, with automatic labels; it cannot score tasks or decisions, which have no objective label in that data.

Statistical validity. All results use one match threshold ($\theta = 0.6$, Definition 4) and one bootstrap configuration (3,000 resamples, case resampling over sessions). The TokenMizer/Graphiti-style/Mem0-style ordering, already within noise at this threshold, has not been checked for stability under a threshold sweep, unlike the product repository’s own single-method evaluation harness, which ships one.

External validity. Every method processes one session’s full transcript in a single pass. None of the actual multi-session consolidation behavior the named systems are partly designed for (Mem0’s cross-conversation fact merging, Graphiti’s graph accretion over weeks of interaction) is exercised here; this is a single-session extraction benchmark only, and no claim in this paper should be read as bearing on cross-session performance. Separately, this paper measures extraction quality — whether the right facts are pulled out of a transcript — and does not measure whether having them improves an LLM’s actual continuation of the session; that is a different experiment, requiring live model calls to judge downstream task success, and is not attempted here.

XIV. CONCLUSION AND FUTURE WORK

This paper presents TokenMizer, a system that addresses LLM context loss by modeling session history as a typed knowledge graph and serializing it into compact, structured resume blocks, and evaluates the released 0.5.4 product on a controlled 100-session, 8-method benchmark with bootstrap confidence intervals throughout. TokenMizer significantly outperforms three naive baselines and two other graph-memory strategies (MemGPT-style, GraphRAG-style), and is statistically indistinguishable from two comparably-designed ones (Graphiti-style, Mem0-style), where the honest result is a tie in both directions rather than a loss or a win, and where TokenMizer’s point estimate now matches Mem0-style’s exactly. Decision and error extraction remain TokenMizer’s most improvable categories, both in absolute terms and relative to the two peer methods it otherwise ties, though a targeted fix

traced to the specific missed and spurious items behind an earlier measurement of this same benchmark (Section XII) narrowed both gaps in this release. Every pattern-matching method tested, independent of implementation, shares a common ceiling of roughly 20% macro F1 on text that states facts indirectly, unmoved by that fix since it addresses vocabulary coverage rather than the underlying pattern-matching strategy.

On 1,467 real agent sessions with automatic ground truth (Section XI), real traffic exposed defects that the labelled corpora had not. Fixing them, with defects located on a development split only, raised runtime-error recall on the held-out split from 63% to 92% and more than doubled the share of runtime errors that reach the resume block, for every model and both agent frameworks. We take two lessons from this for work on session memory. The first is that a memory system should be measured on the traffic it will receive, not only on labelled corpora. The second is that such traffic supplies objective labels for part of what matters, namely edited files and observed failures.

Two directions follow directly from these results. **First**, evaluating the product’s existing LLM extraction path (Section V-D) against the same $n = 100$ corpus would test directly whether a model call closes the register gap of Section X-F, rather than leaving it as the inferred but untested explanation this paper offers. **Second**, the same test would need to be run fairly against LLM-backed versions of the four comparison methods, not against TokenMizer alone, for any resulting claim of superiority to carry the same evidentiary weight as the comparisons in this paper.

DATA AND CODE AVAILABILITY

The 100-session corpus, all eight method implementations, the scoring harness, and raw per-session results (JSON and CSV) for every number in Section X are released at <https://github.com/Shweta-Mishra-ai/tokenmizer-research>. The corpus generator (benchmarks/memorybench/generate.py) is seeded and produces byte-identical output on re-run. The real-agent evaluation (Section XI) is reproducible from public data. benchmarks/realworld/ contains the fetch script, the converters, the frozen protocol and the runner. The raw per-session results for both product commits and both conditions are in benchmarks/results/realworld/. The TokenMizer product itself is released separately at <https://github.com/Shweta-Mishra-ai/tokenmizer> under the MIT licence; this manuscript evaluates release 0.5.4.

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